

A Problem in Mahjong

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Introduction

Once upon a time, the IMO Hong Kong team owned a set of mahjong. The team members would play mahjong for fun during the IMO. They would also collect a coin from the host country and keep it inside the set of mahjong. However, the mahjong together with the coins was lost in Germany in 2009. The new set of mahjong was kept by the team members again since 2014.

There are many mathematical questions in the game of mahjong. In this note, we address one famous counting problem in this game. The question is usually solved by computers, but here we shall provide a mathematical solution. To avoid using any terminologies in the game, we rephrase the whole question in mathematical languages.

Main Problem

Problem 1. There are 136 cards with labels

$$1, 2, \dots, 9, 11, 12, \dots, 19, 21, 22, \dots, 29, 31, 41, 51, 61, 71, 81, 91$$

respectively such that there are exactly 4 cards of each label. A triple of cards is called *nice* if they have the same labels or their labels are consecutive integers. Suppose we randomly pick 14 cards. We can win the game if one of the following conditions holds:

- (i) the 14 cards can be partitioned into 4 nice triples and a pair of cards with the same label; or
- (ii) the labels of all 14 cards are congruent to $\pm 1 \pmod{10}$, and only 2 of them have the same label.

Find the probability that we win the game.

For example, we can win the game in each of the following cases (where the labels are shown).

- 3, 3, 3, 6, 6, 6, 9, 9, 9, 14, 14, 51, 51, 51
- 1, 2, 3, 7, 7, 7, 12, 13, 14, 14, 14, 29, 29, 29
- 2, 2, 2, 3, 4, 4, 4, 5, 5, 5, 6, 6, 7, 7 (which can be partitioned as $\{2, 2, 2\}$, $\{3, 4, 5\}$, $\{5, 6, 7\}$, $\{5, 6, 7\}$, $\{4, 4\}$, for example)
- 1, 9, 9, 11, 19, 21, 29, 31, 41, 51, 61, 71, 81, 91

Full Solution

Solution. The answer is $\frac{11354436764346}{\binom{136}{14}} \approx \frac{1}{374330} \approx 2.67 \times 10^{-6}$

In the following, we only focus on the labels and we assume all cards with the same label are distinct. For example, there are $\binom{4}{2} = 6$ ways to choose exactly 2 copies of 91. We partition all cards into 4 types as follows.

- (Type A) 1, 2, 3, 4, 5, 6, 7, 8, 9
- (Type B) 11, 12, 13, 14, 15, 16, 17, 18, 19
- (Type C) 21, 22, 23, 24, 25, 26, 27, 28, 29
- (Type D) 31, 41, 51, 61, 71, 81, 91

Note that a nice triple must come from the same type. Therefore, it suffices to see how each type can form some nice triples and/or an equal pair.

Case 1. Type D without the equal pair

Clearly, there can be 1, 2, 3, 4 nice triples, and each nice triple must consist of the same label. Note that there are 7 distinct labels in type D.

- The number of ways to choose 1 nice triple is $\binom{7}{1} \times \binom{4}{3} = 28$.
- The number of ways to choose 2 nice triples is $\binom{7}{2} \times \binom{4}{3}^2 = 336$.
- The number of ways to choose 3 nice triples is $\binom{7}{3} \times \binom{4}{3}^3 = 2240$.
- The number of ways to choose 4 nice triples is $\binom{7}{4} \times \binom{4}{3}^4 = 8960$.

Case 2. Type D with an equal pair

Firstly, there are $\binom{7}{1} \times \binom{4}{2} = 42$ ways to choose the equal pair.

- The number of ways to choose 1 nice triple is $42 \times \binom{6}{1} \times \binom{4}{3} = 1008$.
- The number of ways to choose 2 nice triples is $42 \times \binom{6}{2} \times \binom{4}{3}^2 = 10080$.

- The number of ways to choose 3 nice triples is $42 \times \binom{6}{3} \times \binom{4}{3}^3 = 53760$.
- The number of ways to choose 4 nice triples is $42 \times \binom{6}{4} \times \binom{4}{3}^4 = 161280$.

Case 3. Type A without the equal pair

- Suppose there is 1 nice triple.
 - There are $\binom{9}{1} \times \binom{4}{3} = 36$ ways to choose *aaa*.
 - There are $7 \times \binom{4}{1}^3 = 448$ ways to choose *abc* (meaning 3 consecutive labels).

The total number of ways to choose 1 nice triple is $36 + 448 = 484$.

- Suppose there are 2 nice triples.
 - There are $\binom{9}{2} \times \binom{4}{3}^2 = 576$ ways to choose *aaa* and *bbb*.
 - There are $7 \times 3 \times \binom{4}{1}^2 = 336$ ways to choose *abc* and *ddd* such that *d* is one of *a, b, c*.
 - There are $7 \times 6 \times \binom{4}{1}^3 \times \binom{4}{3} = 10752$ ways to choose *abc* and *ddd* such that *d* is not *a, b, c*.
 - There are $7 \times \binom{4}{2}^3 = 1512$ ways to choose *aabbcc* ($\{abc, abc\}$).
 - There are $6 \times \binom{4}{1}^2 \times \binom{4}{2}^2 = 3456$ ways to choose *abbccd* ($\{abc, bcd\}$).
 - There are $5 \times \binom{4}{1}^4 \times \binom{4}{2} = 7680$ ways to choose *abccde* ($\{abc, cde\}$).
 - There are $10 \times \binom{4}{1}^6 = 40960$ ways to choose *abc* and *xyz* such that all of *a, b, c, x, y, z* are distinct. Note that there are 10 cases as listed below.

$$\begin{aligned} &\{123, 456\}, \{123, 567\}, \{123, 678\}, \{123, 789\}, \{234, 567\}, \\ &\{234, 678\}, \{234, 789\}, \{345, 678\}, \{345, 789\}, \{456, 789\} \end{aligned}$$

The total number of ways to choose 2 nice triples is

$$576 + 336 + 10752 + 1512 + 3456 + 7680 + 40960 = 65272.$$

- Suppose there are 3 nice triples.

We count according to the number of cards having the same labels. For example, the case 4311 means there are 4, 3, 1, 1 cards with labels a, b, c, d respectively, where a, b, c, d are distinct (not necessarily consecutive).

- (441) The 3 labels must be consecutive and the order does not matter. For example, $abbbccccc$ can be partitioned as $\{abc, bbb, ccc\}$. The number of ways is $7 \times 3 \times \binom{4}{1} = 84$.
- (432) Impossible.
- (4311) The partition must be $3 + 411$, where the labels in 411 must be consecutive and the order does not matter. For example, $abbbbc$ can be partitioned as $\{abc, bbb\}$. The number of ways is $7 \times 3 \times 6 \times \binom{4}{1}^2 \times \binom{4}{3} = 8064$.
- (4221) The 4 labels must be consecutive and the order can be 4221, 1224. For example, $aaaabbccd$ can be partitioned as $\{aaa, abc, bcd\}$. The number of ways is $6 \times 2 \times \binom{4}{1} \times \binom{4}{2}^2 = 1728$.
- (42111) The 5 labels must be consecutive and the two equal labels must be the middle label. The order of the other labels does not matter. For example, $abbbccde$ can be partitioned as $\{abc, bbb, cde\}$. The number of ways is $5 \times 4 \times \binom{4}{1}^3 \times \binom{4}{2} = 7680$.
- (411111) The partition must be $411 + 111$, where the labels in each group must be consecutive and the order does not matter. For example, $abccccxyz$ can be partitioned as $\{abc, ccc, xyz\}$. As above, there are 10 cases of the form $111 + 111$. Therefore, the number of ways is $10 \times 6 \times \binom{4}{1}^5 = 61440$.
- (333) The labels can be arbitrary as we can partition this case as $3 + 3 + 3$. The number of ways is $\binom{9}{3} \times \binom{4}{3}^3 = 5376$.
- (3321) The 4 labels must be consecutive and the order can be 2331, 1332. For example, $abbbccdd$ can be partitioned as $\{abc, bcd, bcd\}$. The number of ways is $6 \times 2 \times \binom{4}{1} \times \binom{4}{2} \times \binom{4}{3}^2 = 4608$.
- (33111) The partition must be $3 + 3 + 111$, where the labels in 111 must be consecutive. The number of ways is $7 \times \binom{6}{2} \times \binom{4}{1}^3 \times \binom{4}{3}^2 = 107520$.

- (3222) The partition must be $3 + 222$, where the labels in 222 must be consecutive. The number of ways is $7 \times 6 \times \binom{4}{2}^3 \times \binom{4}{3} = 36288$.

- (32211) There are two possible partitions.

The partition can be $3 + 1221$, where the labels in 1221 must be consecutive. There are $6 \times 5 = 30$ ways to choose the two groups.

The order can be $22311, 12321, 11322$. For example, $abbccdde$ can be partitioned as $\{abc, bcd, cde\}$. There are $5 \times 3 = 15$ ways to choose this group.

The number of ways in this case is $(30 + 15) \times \binom{4}{1}^2 \times \binom{4}{2}^2 \times \binom{4}{3} = 103680$.

- (321111) The partition must be $3 + 11211$, where the labels in 11211 are in this consecutive order. The number of ways is $5 \times 4 \times \binom{4}{1}^4 \times \binom{4}{2} \times \binom{4}{3} = 122880$.

- (3111111) The partition must be $3 + 111 + 111$, where the labels in 111 must be consecutive. The number of ways is $10 \times 3 \times \binom{4}{1}^6 \times \binom{4}{3} = 491520$.

- (22221) Impossible.

- (222111) There are two possible partitions.

The partition can be $222 + 111$, where the labels in each group must be consecutive. There are $10 \times 2 = 20$ ways to choose the two groups.

The order can be $122211, 112221$. For example, $abbccdde$ can be partitioned as $\{abc, bcd, def\}$. There are $4 \times 2 = 8$ ways to choose this group.

The number of ways in this case is $(20 + 8) \times \binom{4}{1}^3 \times \binom{4}{2}^3 = 387072$.

- (2211111) There are two possible partitions.

The partition can be $1221 + 111$, where the labels in each group are in these consecutive orders. There are 12 ways to choose the two groups. We list 6 cases below, and the other 6 cases are obtained via the symmetry $x \mapsto (10 - x)$.

$$\{122334, 567\}, \{122334, 678\}, \{122334, 789\}, \\ \{233445, 678\}, \{233445, 789\}, \{344556, 789\}$$

The order can be 1121211 . $abccdeefg$ can be partitioned as $\{abc, cde, efg\}$. There are 3 ways to choose this group.

The number of ways in this case is $(12 + 3) \times \binom{4}{1}^5 \times \binom{4}{2}^2 = 552960$.

- (21111111) The partition must be $11211 + 111$, where the labels in each group are in these consecutive orders. There are 6 ways to choose the two groups, including 3 cases listed below and 3 other symmetric cases.

$$\{123345, 678\}, \{123345, 789\}, \{234456, 789\}$$

The number of ways is $6 \times \binom{4}{1}^7 \times \binom{4}{2} = 589824$.

- (111111111) There is only one such case and it is valid, since the partition is $\{123, 456, 789\}$. The number of ways is $\binom{4}{1}^9 = 262144$.

The total number of ways to choose 3 nice triples is

$$\begin{aligned} &84 + 8064 + 1728 + 7680 + 61440 + 5376 + 4608 + 107520 + 36288 + 103680 \\ &+ 122880 + 491520 + 387072 + 552960 + 589824 + 262144 \\ &= 2742868. \end{aligned}$$

• Suppose there are 4 nice triples.

- (444) The 3 labels must be consecutive. For example, $aaaabbbbcccc$ can be partitioned as $\{abc, abc, abc, abc\}$. The number of ways is 7.
- (4431) The partition must be $3+441$, where the labels in 441 must be consecutive and the order does not matter. The partition $\{abc, abc, abc, bcd\}$ needs not be considered since this can also be partitioned as $\{aaa, bbb, bcd, ccc\}$. The number of ways is $7 \times 3 \times 6 \times \binom{4}{1} \times \binom{4}{3} = 2016$.
- (4422) The 4 labels must be consecutive and the order can be 4224, 2442. For example, $aabbbbccccdd$ can be partitioned as $\{abc, abc, bcd, bcd\}$. The number of ways is $6 \times 2 \times \binom{4}{2}^2 = 432$.
- (44211) The order must be 11211 with two 1's being replaced by 4's. The number of ways is $5 \times \binom{4}{2} \times \binom{4}{1}^2 \times \binom{4}{2} = 2880$.
- (441111) The partition must be $111 + 111$ with two 1's being replaced by 4's. The number of ways is $10 \times \binom{6}{2} \times \binom{4}{1}^4 = 38400$.

- (4332) The 4 labels must be consecutive and the order can be 4332, 2334. For example, *aaaabbbcccd* can be partitioned as $\{aaa, abc, bcd, bcd\}$. The number of ways is $6 \times 2 \times \binom{4}{2} \times \binom{4}{3}^2 = 1152$.
- (43311) There are two possible partitions.
 The partition can be 3+3+411, where the labels in 411 must be consecutive and the order does not matter. There are $7 \times 3 \times \binom{6}{2} = 315$ ways to choose the three groups.
 The order can be 13431. *abbbccccddde* can be partitioned as $\{abc, bcd, bcd, cde\}$. There are 5 ways to choose this group.
 The number of ways in this case is $(315 + 5) \times \binom{4}{1}^2 \times \binom{4}{3}^2 = 81920$.
- (43221) There are three possible partitions.
 The partition can be 3+4221, where the labels in 4221 must be consecutive and the order can be 4221, 1224. There are $6 \times 2 \times 5 = 60$ ways to choose the two groups.
 The order can be 22311, 12321, 11322 with a 1 being replaced by 4. There are $5 \times 2 \times 3 = 30$ ways to choose this group.
 The order can be 23421, 12432. For example, *aabbbccccdde* can be partitioned as $\{abc, abc, bcd, cde\}$. There are $5 \times 2 = 10$ ways to choose this group.
 The number of ways in this case is $(60 + 30 + 10) \times \binom{4}{1} \times \binom{4}{2}^2 \times \binom{4}{3} = 57600$.
- (432111) The partition must be 3+11211 with a 1 being replaced by 4. The number of ways is $5 \times 4 \times 4 \times \binom{4}{1}^3 \times \binom{4}{2} \times \binom{4}{3} = 122880$.
- (4311111) The partition must be 3+111+111 with a 1 being replaced by 4. The number of ways is $10 \times 6 \times 3 \times \binom{4}{1}^5 \times \binom{4}{3} = 737280$.
- (42222) The 5 labels must be consecutive and the order must be 22422. Indeed, *aabbbccccdee* can be partitioned as $\{abc, abc, cde, cde\}$. The number of ways is $5 \times \binom{4}{2}^4 = 6480$.

- (422211) There are two possible partitions.

The partition can be $222 + 411$, where the labels in each group must be consecutive and the order does not matter. There are $10 \times 2 \times 3 = 60$ ways to choose the two groups.

The order can be $122211, 112221$ with a 1 being replaced by 4. There are $4 \times 3 \times 2 = 24$ ways to choose this group.

The number of ways in this case is $(60 + 24) \times \binom{4}{1}^2 \times \binom{4}{2}^3 = 290304$.

- (4221111) There are two possible partitions.

The partition can be $1221 + 111$ with a 1 being replaced by 4. There are $12 \times 5 = 60$ ways to choose the two groups.

The order can be 1121211 with a 1 being replaced by 4. There are $3 \times 5 = 15$ ways to choose this group.

The number of ways in this case is $(60 + 15) \times \binom{4}{1}^4 \times \binom{4}{2}^2 = 691200$.

- (42111111) The partition must be $11211 + 111$ with a 1 being replaced by 4. The number of ways is $6 \times 7 \times \binom{4}{1}^6 \times \binom{4}{2} = 1032192$.

- (411111111) The order must be 111111111 with a 1 being replaced by 4. The number of ways is $9 \times \binom{4}{1}^8 = 589824$.

- (3333) The labels can be arbitrary as we can partition this case as $3+3+3+3$. The number of ways is $\binom{9}{4} \times \binom{4}{3}^4 = 32256$.

- (33321) The partition must be $3 + 2331$, where the labels in 2331 must be consecutive and the order can be $2331, 1332$. The number of ways is $6 \times 2 \times 5 \times \binom{4}{1} \times \binom{4}{2} \times \binom{4}{3}^3 = 92160$.

- (333111) There are two possible partitions.

The partition can be $3 + 3 + 3 + 111$, where the labels in 111 must be consecutive. There are $7 \times \binom{6}{3} = 140$ ways to choose the four groups.

The order can be $133311, 113331$. For example, $abcccddeef$ can be partitioned as $\{abc, cde, cde, def\}$. There are $4 \times 2 = 8$ ways to choose this group.

The number of ways in this case is $(140 + 8) \times \binom{4}{1}^3 \times \binom{4}{3}^3 = 606208$.

- (33222) The partition must be $3 + 3 + 222$, where the labels in 222 must be consecutive. The number of ways is $7 \times \binom{6}{2} \times \binom{4}{2}^3 \times \binom{4}{3}^2 = 362880$.

- (332211) There are three possible partitions.

The partition can be $3 + 3 + 1221$, where the labels in 1221 must be consecutive. There are $6 \times \binom{5}{2} = 60$ ways to choose the three groups.

The partition can be $3 + 22311$, where the labels in 22311 must be consecutive and the order can be $22311, 12321, 11322$. There are $5 \times 3 \times 4 = 60$ ways to choose the two groups.

The order can be $233211, 123321, 112332$. For example, $aabbcccddef$ can be partitioned as $\{abc, abc, bcd, def\}$. There are $4 \times 3 = 12$ ways to choose this group.

The number of ways in this case is $(60 + 60 + 12) \times \binom{4}{1}^2 \times \binom{4}{2}^2 \times \binom{4}{3}^2 = 1216512$.

- (3321111) There are three possible partitions.

The partition can be $3 + 3 + 11211$, where the labels in 11211 must be consecutive. There are $5 \times \binom{4}{2} = 30$ ways to choose the three groups.

The partition can be $2331 + 111$, where the labels in each group are consecutive and the order in 2331 can be $2331, 1332$. There are $12 \times 2 = 24$ ways to choose the two groups.

The order can be 1132311 . $abcccddeeffg$ can be partitioned as $\{abc, cde, cde, efg\}$. There are 3 ways to choose this group.

The number of ways in this case is $(30 + 24 + 3) \times \binom{4}{1}^4 \times \binom{4}{2} \times \binom{4}{3}^2 = 1400832$.

- (33111111) The partition must be $3 + 3 + 111 + 111$, where the labels in 111 must be consecutive. The number of ways is $10 \times \binom{3}{2} \times \binom{4}{1}^6 \times \binom{4}{3}^2 = 1966080$.

- (322221) The order must be $223221, 122322$. For example, $aabbcccddeef$ can be partitioned as $\{abc, abc, cde, def\}$. Thus, the number of ways is $4 \times 2 \times \binom{4}{1} \times \binom{4}{2}^4 \times \binom{4}{3} = 165888$.

- (3222111) There are three possible partitions.

The partition can be $3 + 222 + 111$, where the labels in each group must be consecutive. There are $10 \times 2 \times 3 = 60$ ways to choose the three groups.

The partition can be $3 + 122211$, where the labels in 122211 must be consecutive and the order can be $122211, 112221$. There are $4 \times 2 \times 3 = 24$ ways to choose the two groups.

The order can be $2231211, 1232211, 1122321, 1121322$. For example, the case $aabbccddeeffg$ can be partitioned as $\{abc, abc, cde, efg\}$. There are $3 \times 4 = 12$ ways to choose this group.

The number of ways in this case is $(60+24+12) \times \binom{4}{1}^3 \times \binom{4}{2}^3 \times \binom{4}{3} = 5308416$.

- (32211111) There are three possible partitions.

The partition can be $3 + 1221 + 111$, where the labels in each group are in these consecutive orders. There are $12 \times 2 = 24$ ways to choose the three groups.

The partition can be $3 + 1121211$, where the labels in 1121211 are in this consecutive order. There are $3 \times 2 = 6$ ways to choose the two groups.

The partition can be $22311 + 111$, where the labels in each group are consecutive and the order in 22311 can be $22311, 12321, 11322$. There are $6 \times 3 = 18$ ways to choose the two groups.

The number of ways in this case is $(24+6+18) \times \binom{4}{1}^5 \times \binom{4}{2}^2 \times \binom{4}{3} = 7077888$.

- (321111111) The partition must be $3 + 11211 + 111$, where the labels in each group are in these consecutive orders. Note that we only need to permute the order of the three groups as there are 9 labels in total. The number of ways is $3! \times \binom{4}{1}^7 \times \binom{4}{2} \times \binom{4}{3} = 2359296$.

- (222222) The partition must be $222 + 222$, where the labels in 222 must be consecutive. The number of ways is $10 \times \binom{4}{2}^6 = 466560$.

- (2222211) There are two possible partitions.

The partition can be $1221 + 222$, where the labels in each group are in these consecutive orders. There are 12 ways to choose the two groups.

The order can be 1222221 . Indeed, $abbccddeeffg$ can be partitioned as $\{abc, bcd, def, efg\}$. There are 3 ways to choose this group.

The number of ways in this case is $(12 + 3) \times \binom{4}{1}^2 \times \binom{4}{2}^5 = 1866240$.

- (22221111) There are three possible partitions.

The partition can be 11211 + 222, where the labels in each group are in these consecutive orders. There are 6 ways to choose the two groups.

The partition can be 1221 + 1221, where the labels in 1221 are in this consecutive order. There are 3 ways to choose the two groups.

The order can be 12221211, 11222211, 11212221. For example, the case *abccddeeffgh* can be partitioned as $\{abc, cde, def, fgh\}$. There are $2 \times 3 = 6$ ways to choose this group.

The number of ways in this case is $(6 + 3 + 6) \times \binom{4}{1}^4 \times \binom{4}{2}^4 = 4976640$.

- (22211111) There are three possible partitions.

The three 2's are in consecutive positions. For example, *abccddeeffghi* can be partitioned as $\{abc, bcd, def, ghi\}$. There are 7 ways to choose their positions.

The partition can be 11211 + 1221, where the labels in each group are in these consecutive orders. There are 2 ways to choose the two groups.

The order can be 112121211. There is 1 way to choose this group.

The number of ways in this case is $(7 + 2 + 1) \times \binom{4}{1}^6 \times \binom{4}{2}^3 = 8847360$.

The total number of ways to choose 4 nice triples is

$$\begin{aligned}
& 7 + 2016 + 432 + 2880 + 38400 + 1152 + 81920 + 57600 + 122880 + 737280 \\
& + 6480 + 290304 + 691200 + 1032192 + 589824 + 32256 + 92160 + 606208 \\
& + 362880 + 1216512 + 1400832 + 1966080 + 165888 + 5308416 + 7077888 \\
& + 2359296 + 466560 + 1866240 + 4976640 + 8847360 \\
& = 40399783.
\end{aligned}$$

Case 4. Type A with an equal pair

- Suppose there are 0 nice triples with an equal pair.

There are $\binom{9}{1} \times \binom{4}{2} = 54$ ways to choose the equal pair.

- Suppose there is 1 nice triple with an equal pair.

- There are $9 \times 8 \times \binom{4}{2} \times \binom{4}{3} = 1728$ ways to choose aaa and bb .
- There are $7 \times 3 \times \binom{4}{1}^2 \times \binom{4}{3} = 1344$ ways to choose abc and dd such that d is one of a, b, c .
- There are $7 \times 6 \times \binom{4}{1}^3 \times \binom{4}{2} = 16128$ ways to choose abc and dd such that d is not a, b, c .

The total number of ways to choose 1 nice triple with an equal pair is

$$1728 + 1344 + 16128 = 19200.$$

- Suppose there are 2 nice triples with an equal pair.

- (44) Impossible.
- (431) The 3 labels must be consecutive and the order does not matter. For example, $abbbbccc$ can be partitioned as $\{abc, bbb, cc\}$. The number of ways is $7 \times 3! \times \binom{4}{1} \times \binom{4}{3} = 672$.
- (422) The 3 labels must be consecutive and the order does not matter. For example, $aaaabbcc$ can be partitioned as $\{abc, abc, aa\}$. The number of ways is $7 \times 3 \times \binom{4}{2}^2 = 756$.
- (4211) There are two possible partitions.

The partition can be $2 + 411$, where the labels in 411 must be consecutive and the order does not matter. There are $7 \times 3 \times 6 = 126$ ways to choose the two groups.

The order can be $1421, 1241$. For example, $abbbbccd$ can be partitioned as $\{abc, bcd, bb\}$. There are $6 \times 2 = 12$ ways to choose this group.

The number of ways in this case is $(126 + 12) \times \binom{4}{1}^2 \times \binom{4}{2} = 13248$.

- (41111) The order must be 11411 . Indeed, $abccccde$ can be partitioned as $\{abc, cde, cc\}$. The number of ways is $5 \times \binom{4}{1}^4 = 1280$.

- (332) The labels can be arbitrary as we can partition this case as $3 + 3 + 2$. The number of ways is $\binom{9}{2} \times 7 \times \binom{4}{2} \times \binom{4}{3}^2 = 24192$.

- (3311) The partition must be $3 + 311$, where the labels in 311 must be consecutive and the order does not matter. For example, *abbbc* can be partitioned as $\{abc, bb\}$. There are $7 \times 3 \times 6 = 126$ ways to choose the two groups.

However, the order 3113 is counted twice. There are 6 ways to choose this group.

The number of ways in this case is $(126 - 6) \times \binom{4}{1}^2 \times \binom{4}{3}^2 = 30720$.

- (3221) The 4 labels must be consecutive and the order can be 3221, 1223. For example, *aaabbccd* can be partitioned as $\{abc, bcd, aa\}$. The number of ways is $6 \times 2 \times \binom{4}{1} \times \binom{4}{2}^2 \times \binom{4}{3} = 6912$.

- (32111) There are two possible partitions.

The partition can be $3 + 2 + 111$, where the labels in 111 must be consecutive. There are $7 \times 6 \times 5 = 210$ ways to choose the three groups.

The order can be 11211 with a 1 being replaced by 3. There are $5 \times 4 = 20$ ways to choose this group.

The number of ways in this case is $(210 + 20) \times \binom{4}{1}^3 \times \binom{4}{2} \times \binom{4}{3} = 353280$.

- (311111) The partition must be $311 + 111$, where the labels in each group must be consecutive and the order does not matter. The number of ways is $10 \times 6 \times \binom{4}{1}^5 \times \binom{4}{3} = 245760$.

- (2222) The partition must be $2 + 222$, where the labels in 222 must be consecutive. There are $7 \times 6 = 42$ ways to choose the two groups.

However, the order 2222 is counted twice. There are 6 ways to choose this group.

The number of ways in this case is $(42 - 6) \times \binom{4}{2}^4 = 46656$.

- (22211) The partition must be $2 + 1221$, where the labels in 1221 are in this consecutive order. The number of ways is $6 \times 5 \times \binom{4}{1}^2 \times \binom{4}{2}^3 = 103680$.

- (221111) The partition must be $2 + 11211$, where the labels in 11211 are in this consecutive order. The number of ways is $5 \times 4 \times \binom{4}{1}^4 \times \binom{4}{2}^2 = 184320$.
- (2111111) The partition must be $2 + 111 + 111$, where the labels in 111 must be consecutive. The number of ways is $10 \times 3 \times \binom{4}{1}^6 \times \binom{4}{2} = 737280$.
- (11111111) Impossible.

The total number of ways to choose 2 nice triples with an equal pair is

$$\begin{aligned}
& 672 + 756 + 13248 + 1280 + 24192 + 30720 + 6912 + 353280 + 245760 + 46656 \\
& + 103680 + 184320 + 737280 \\
& = 1748756.
\end{aligned}$$

- Suppose there are 3 nice triples with an equal pair.

- (443) The 3 labels must be consecutive and the order does not matter. For example, $aaaabbbbccc$ can be partitioned as $\{aaa, abc, bbb, cc\}$. The number of ways is $7 \times 3 \times \binom{4}{3} = 84$.

- (4421) There are two possible partitions.

The partition can be $2 + 441$, where the labels in 441 must be consecutive and the order does not matter. There are $7 \times 3 \times 6 = 126$ ways to choose the two groups.

The order can be $4221, 1224$ with a 2 being replaced by 4. Thus, there are $6 \times 2 \times 2 = 24$ ways to choose this group.

The number of ways in this case is $(126 + 24) \times \binom{4}{1} \times \binom{4}{2} = 3600$.

- (44111) The order must be 11411 with a 1 being replaced by 4. The number of ways is $5 \times 4 \times \binom{4}{1}^3 = 1280$.

- (4331) There are two possible partitions.

The partition can be $3 + 431$, where the labels in 431 must be consecutive and the order does not matter. There are $7 \times 3! \times 6 = 252$ ways to choose the two groups.

However, the orders $3413, 3143$ are counted twice. There are $6 \times 2 = 12$ ways to choose this group.

The order can be 4331, 1334. For example, *aaaabbbcccd* can be partitioned as $\{abc, abc, bcd, aa\}$. There are $6 \times 2 = 12$ ways to choose this group.

The number of ways in this case is $(252 - 12 + 12) \times \binom{4}{1} \times \binom{4}{3}^2 = 16128$.

- (4322) The partition must be 3 + 422, where the labels in 422 must be consecutive and the order does not matter. Thus, the number of ways is $7 \times 3 \times 6 \times \binom{4}{2}^2 \times \binom{4}{3} = 18144$.

- (43211) There are four possible partitions.

The partition can be 3+2+411, where the labels in 411 must be consecutive and the order does not matter. There are $7 \times 3 \times 6 \times 5 = 630$ ways to choose the three groups.

The order can be 11211 with a 1 being replaced by 3 and a 1 being replaced by 4. There are $5 \times 4 \times 3 = 60$ ways to choose this group.

The partition can be 3 + 1221 with a 2 being replaced by 4. There are $6 \times 2 \times 5 = 60$ ways to choose the two groups.

However, the orders 31241, 14213 are counted twice. There are $5 \times 2 = 10$ ways to choose this group.

The order can be 22311, 12321, 11322 with a 2 being replaced by 4. There are $5 \times 2 \times 3 = 30$ ways to choose this group.

The number of ways in this case is

$$(630 + 60 + 60 - 10 + 30) \times \binom{4}{1}^2 \times \binom{4}{2} \times \binom{4}{3} = 295680.$$

- (431111) There are two possible partitions.

The partition can be 111 + 111 with a 1 being replaced by 3 and a 1 being replaced by 4. There are $10 \times 6 \times 5 = 300$ ways to choose the two groups.

The partition can be 3 + 11411, where the labels in 11411 are in this consecutive order. There are $5 \times 4 = 20$ ways to choose the two groups.

However, the orders 311411, 114113 are counted twice. There are $4 \times 2 = 8$ ways to choose this group.

The number of ways in this case is $(300 + 20 - 8) \times \binom{4}{1}^4 \times \binom{4}{3} = 319488$.

- (42221) The partition must be $2 + 4221$, where the labels in 4221 must be consecutive and the order can be 4221, 1224. The number of ways is $6 \times 2 \times 5 \times \binom{4}{1} \times \binom{4}{2}^3 = 51840$.

- (422111) There are three possible partitions.

The partition can be $222 + 111$ with a 2 being replaced by 4. There are $10 \times 2 \times 3 = 60$ ways to choose the two groups.

The partition can be $2 + 11211$ with a 1 being replaced by 4. There are $5 \times 4 \times 4 = 80$ ways to choose the two groups.

The order can be 122211, 112221 with a 2 being replaced by 4. There are $4 \times 3 \times 2 = 24$ ways to choose this group.

The number of ways in this case is $(60 + 80 + 24) \times \binom{4}{1}^3 \times \binom{4}{2}^2 = 377856$.

- (4211111) There are three possible partitions.

The partition can be $2 + 411 + 111$, where the labels in each group must be consecutive and the order does not matter. There are $10 \times 6 \times 3 = 180$ ways to choose the three groups.

The partition can be $1221 + 111$ with a 2 being replaced by 4. There are $12 \times 2 = 24$ ways to choose the two groups.

The order can be 1121211 with a 2 being replaced by 4. There are $3 \times 2 = 6$ ways to choose this group.

The number of ways in this case is $(180 + 24 + 6) \times \binom{4}{1}^5 \times \binom{4}{2} = 1290240$.

- (41111111) The partition must be $11411 + 111$, where the labels in each group are in these consecutive orders. The number of ways is $6 \times \binom{4}{1}^7 = 98304$.

- (3332) The labels can be arbitrary as we can partition this case as $3 + 3 + 3 + 2$. The number of ways is $\binom{9}{3} \times 6 \times \binom{4}{2} \times \binom{4}{3}^3 = 193536$.

- (33311) The partition must be $3 + 3 + 311$, where the labels in 311 must be consecutive and the order does not matter. There are $7 \times 3 \times \binom{6}{2} = 315$ ways to choose the three groups.

However, the partition $3 + 3113$ is counted twice. There are $6 \times 5 = 30$ ways to choose the two groups.

The number of ways in this case is $(315 - 30) \times \binom{4}{1}^2 \times \binom{4}{3}^3 = 291840$.

- (33221) There are three possible partitions.

The partition can be $3 + 3221$, where the labels in 3221 must be consecutive and the order can be $3221, 1223$. There are $6 \times 2 \times 5 = 60$ ways to choose the two groups.

The partition can be $2 + 2331$, where the labels in 2331 must be consecutive and the order can be $2331, 1332$. There are $6 \times 2 \times 5 = 60$ ways to choose the two groups.

The order can be $22311, 12321, 11322$ with a 1 being replaced by 3 . There are $5 \times 2 \times 3 = 30$ ways to choose this group.

However, the orders $22331, 13322$ are counted twice. There are $5 \times 2 = 10$ ways to choose this group.

The number of ways in this case is

$$(60 + 60 + 30 - 10) \times \binom{4}{1} \times \binom{4}{2}^2 \times \binom{4}{3}^2 = 322560.$$

- (332111) There are two possible partitions.

The partition can be $3 + 3 + 2 + 111$, where the labels in 111 must be consecutive. There are $7 \times \binom{6}{2} \times 4 = 420$ ways to choose the four groups.

The partition can be $3 + 11211$ with a 1 being replaced by 3 . There are $5 \times 4 \times 4 = 80$ ways to choose the two groups.

The number of ways in this case is $(420 + 80) \times \binom{4}{1}^3 \times \binom{4}{2} \times \binom{4}{3}^2 = 3072000$.

- (3311111) The partition must be $3 + 111 + 111$ with a 1 being replaced by 3 . There are $10 \times 6 \times 3 = 180$ ways to choose the three groups.

However, the partition $3113 + 111$ and the order 3111113 are counted twice. There are $12 + 3 = 15$ ways to choose the two groups.

The number of ways in this case is $(180 - 15) \times \binom{4}{1}^5 \times \binom{4}{3}^2 = 2703360$.

- (32222) The partition must be $3 + 2 + 222$, where the labels in 222 must be consecutive. There are $7 \times 6 \times 5 = 210$ ways to choose the three groups.

However, the partition $3 + 2222$ is counted twice. There are $6 \times 5 = 30$ ways to choose the two groups.

The number of ways in this case is $(210 - 30) \times \binom{4}{2}^4 \times \binom{4}{3} = 933120$.

- (322211) There are four possible partitions.

The partition can be $3 + 2 + 1221$, where the labels in 1221 are in this consecutive order. There are $6 \times 5 \times 4 = 120$ ways to choose the three groups.

The partition can be $2 + 22311$, where the labels in 22311 must be consecutive and the order can be $22311, 12321, 11322$. There are $5 \times 3 \times 4 = 60$ ways to choose the two groups.

The partition can be $222 + 111$ with a 1 being replaced by 3 . There are $10 \times 2 \times 3 = 60$ ways to choose the two groups.

However, the orders $222311, 113222$ are counted twice. There are $4 \times 2 = 8$ ways to choose this group.

The order can be $122211, 112221$ with a 1 being replaced by 3 . There are $4 \times 3 \times 2 = 24$ ways to choose this group.

The number of ways in this case is

$$(120 + 60 + 60 - 8 + 24) \times \binom{4}{1}^2 \times \binom{4}{2}^3 \times \binom{4}{3} = 3538944.$$

- (3221111) There are three possible partitions.

The partition can be $3 + 2 + 11211$, where the labels in 11211 are in this consecutive order. There are $5 \times 4 \times 3 = 60$ ways to choose the three groups.

The partition can be $1221 + 111$ with a 1 being replaced by 3 . There are $12 \times 5 = 60$ ways to choose the two groups.

The order can be 1121211 with a 1 being replaced by 3 . There are $3 \times 5 = 15$ ways to choose this group.

The number of ways in this case is $(60 + 60 + 15) \times \binom{4}{1}^4 \times \binom{4}{2}^2 \times \binom{4}{3} = 4976640$.

- (32111111) There are two possible partitions.

The partition can be $3 + 2 + 111 + 111$, where the labels in 111 must be consecutive. There are $10 \times 3 \times 2 = 60$ ways to choose the four groups.

The partition can be $11211 + 111$ with a 1 being replaced by 3. There are $6 \times 7 = 42$ ways to choose the two groups.

The number of ways in this case is $(60 + 42) \times \binom{4}{1}^6 \times \binom{4}{2} \times \binom{4}{3} = 10027008$.

- (311111111) The order must be 111111111 with a 1 being replaced by 3. The number of ways is $9 \times \binom{4}{1}^8 \times \binom{4}{3} = 2359296$.

- (222221) Impossible.

- (2222111) There are two possible partitions.

The partition can be $2 + 222 + 111$, where the labels in each group must be consecutive. There are $10 \times 2 \times 3 = 60$ ways to choose the three groups.

However, the partition $2222 + 111$ is counted twice. There are 12 ways to choose the two groups.

The partition can be $2 + 122211$, where the labels in 122211 must be consecutive and the order can be 122211, 112221. There are $4 \times 2 \times 3 = 24$ ways to choose the two groups.

The number of ways in this case is $(60 - 12 + 24) \times \binom{4}{1}^3 \times \binom{4}{2}^4 = 5971968$.

- (222111111) There are two possible partitions.

The partition can be $2 + 1221 + 111$, where the labels in each group are in these consecutive orders. There are $12 \times 2 = 24$ ways to choose the three groups.

The partition can be $2 + 1121211$ where the labels in 1121211 are in this consecutive order. There are $3 \times 2 = 6$ ways to choose the two groups.

The number of ways in this case is $(24 + 6) \times \binom{4}{1}^5 \times \binom{4}{2}^3 = 6635520$.

- (221111111) The partition must be $2 + 11211 + 111$, where the labels in each group are in these consecutive orders. Thus, the number of ways is $3! \times \binom{4}{1}^7 \times \binom{4}{2}^2 = 3538944$.

The total number of ways to choose 3 nice triples with an equal pair is

$$\begin{aligned}
& 84 + 3600 + 1280 + 16128 + 18144 + 295680 + 319488 + 51840 + 377856 \\
& + 1290240 + 98304 + 193536 + 291840 + 322560 + 3072000 + 2703360 \\
& + 933120 + 3538944 + 4976640 + 10027008 + 2359296 + 5971968 \\
& + 6635520 + 3538944 \\
& = 47037380.
\end{aligned}$$

- Suppose there are 4 nice triples.

- (4442) There are two possible partitions.

The partition can be $2 + 444$, where the labels in 444 must be consecutive. There are $7 \times 6 = 42$ ways to choose the two groups.

The order can be 4424, 4244. For example, *aaaabbbbccdddd* can be partitioned as $\{aaa, abc, bcd, ddd, bb\}$. There are $6 \times 2 = 12$ ways to choose this group.

The number of ways in this case is $(42 + 12) \times \binom{4}{2} = 324$.

- (44411) The order must be 11411 with two 1's being replaced by 4's. The number of ways is $5 \times \binom{4}{2} \times \binom{4}{1}^2 = 480$.

- (4433) There are two possible partitions.

The partition can be $3 + 443$, where the labels in 443 must be consecutive and the order does not matter. There are $7 \times 3 \times 6 = 126$ ways to choose the two groups.

However, the order 3443 is counted twice. There are 6 ways to choose this group.

The order can be 4334. Indeed, *aaaabbbbccddddd* can be partitioned as $\{aaa, abc, bcd, bcd, dd\}$. There are 6 ways to choose this group.

The number of ways in this case is $(126 - 6 + 6) \times \binom{4}{3}^2 = 2016$.

- (44321) There are five possible partitions.

The partition can be $3 + 2 + 441$, where the labels in 441 must be consecutive and the order does not matter. There are $7 \times 3 \times 6 \times 5 = 630$ ways to choose the three groups.

The partition can be $3 + 4421$, where the labels in 4421 must be consecutive and the order can be $4421, 4241, 1424, 1244$. There are $6 \times 4 \times 5 = 120$ ways to choose the two groups.

The order can be 11211 with a 1 being replaced by 3 and two 1 's being replaced by 4 's. There are $5 \times 4 \times \binom{3}{2} = 60$ ways to choose this group.

However, the orders $44213, 34241, 31244, 14243$ are counted twice. There are $5 \times 4 = 20$ ways to choose this group.

The order can be $22311, 12321, 11322$ with a 1 being replaced by 4 and a 2 being replaced by 4 . There are $5 \times 2 \times 2 \times 3 = 60$ ways to choose this group.

However, the orders $42341, 14324$ are counted twice. There are $5 \times 2 = 10$ ways to choose this group.

The order can be $43421, 12434$. For example, $aaaabbbccccdde$ can be partitioned as $\{abc, abc, bcd, cde, aa\}$. There are $5 \times 2 = 10$ ways to choose this group.

The number of ways in this case is

$$(630 + 120 + 60 - 20 + 60 - 10 + 10) \times \binom{4}{1} \times \binom{4}{2} \times \binom{4}{3} = 81600.$$

– (443111) There are two possible partitions.

The partition can be $111 + 111$ with a 1 being replaced by 3 and two 1 's being replaced by 4 . There are $10 \times 6 \times \binom{5}{2} = 600$ ways to choose the two groups.

The partition can be $3 + 11411$ with a 1 being replaced by 4 . There are $5 \times 4 \times 4 = 80$ ways to choose the two groups.

However, the orders $311411, 114113$ with a 1 being replaced by 4 are counted twice. There are $4 \times 4 \times 2 = 32$ ways to choose this group.

The number of ways in this case is $(600 + 80 - 32) \times \binom{4}{1}^3 \times \binom{4}{3} = 165888$.

– (44222) There are two possible partitions.

The partition can be $2 + 4224$, where the labels in 4224 must be consecutive and the order can be $4224, 2442$. There are $6 \times 2 \times 5 = 60$ ways to choose the two groups.

The order can be 42422, 22424. For example, *aaaabbccccddde* can be partitioned as $\{abc, abc, cde, cde, aa\}$. There are $5 \times 2 = 10$ ways to choose this group.

The number of ways in this case is $(60 + 10) \times \binom{4}{2}^3 = 15120$.

- (442211) There are three possible partitions.

The partition can be $2 + 11211$ with two 1's being replaced by 4's. There are $5 \times \binom{4}{2} \times 4 = 120$ ways to choose the two groups.

The partition can be $422 + 411$, where the labels in each group must be consecutive and the order does not matter. There are $10 \times 2 \times 3 \times 3 = 180$ ways to choose the two groups.

The order can be 122211, 112221 with a 1 being replaced by 4 and a 2 being replaced by 4. There are $4 \times 3 \times 3 \times 2 = 72$ ways to choose this group.

However, the orders 422411, 142241, 114112 are counted twice. There are $4 \times 3 = 12$ ways to choose this group.

The number of ways in this case is $(120 + 180 + 72 - 12) \times \binom{4}{1}^2 \times \binom{4}{2}^2 = 207360$.

- (4421111) There are three possible partitions.

The partition can be $2 + 111 + 111$ with two 1's being replaced by 4. There are $10 \times \binom{6}{2} \times 3 = 450$ ways to choose the three groups.

The partition can be $1221 + 111$ with a 1 being replaced by 4 and a 2 being replaced by 4. There are $12 \times 5 \times 2 = 120$ ways to choose the two groups.

The order can be 1121211 with a 1 being replaced by 4 and a 2 being replaced by 4. There are $3 \times 5 \times 2 = 30$ ways to choose this group.

However, the orders 1421411, 1141241 are counted twice. There are $3 \times 2 = 6$ ways to choose this group.

The number of ways in this case is $(450 + 120 + 30 - 6) \times \binom{4}{1}^4 \times \binom{4}{2} = 912384$.

- (44111111) The partition must be $11411 + 111$ with a 1 being replaced by 4. There are $6 \times 7 = 42$ ways to choose the two groups.

However, the order 11411411 is counted twice. There are 2 ways to choose this group.

The number of ways in this case is $(42 - 2) \times \binom{4}{1}^6 = 163840$.

- (43331) There are two possible partitions.

The partition can be $3+3+431$, where the labels in 431 must be consecutive and the order does not matter. There are $7 \times 3! \times \binom{6}{2} = 630$ ways to choose the three groups.

However, the partition $3 + 3413$ is counted twice, where the labels in 3413 must be consecutive and the order can be $3413, 3143$. There are $6 \times 2 \times 5 = 60$ ways to choose the two groups.

The partition can be $3+4331$, where the labels in 4331 must be consecutive and the order can be $4331, 1334$. There are $6 \times 2 \times 5 = 60$ ways to choose the two groups.

The number of ways in this case is $(630 - 60 + 60) \times \binom{4}{1} \times \binom{4}{3}^3 = 161280$.

- (43322) There are four possible partitions.

The partition can be $3+3+422$, where the labels in 422 must be consecutive and the order does not matter. There are $7 \times 3 \times \binom{6}{2} = 315$ ways to choose the three groups.

The partition can be $2+4332$, where the labels in 4332 must be consecutive and the order can be $4332, 2334$. There are $6 \times 2 \times 5 = 60$ ways to choose the two groups.

The order can be $22311, 12321, 11322$ with a 1 being replaced by 3 and a 1 being replaced by 4 . There are $5 \times 2 \times 3 = 30$ ways to choose this group.

However, the orders $43322, 22334$ are counted twice. There are $5 \times 2 = 10$ ways to choose this group.

The order can be $23423, 32432$. For example, $aabbbccccddeee$ can be partitioned as $\{abc, abc, bcd, cde, ee\}$. There are $5 \times 2 = 10$ ways to choose this group.

The number of ways in this case is

$$(315 + 60 + 30 - 10 + 10) \times \binom{4}{2}^2 \times \binom{4}{3}^2 = 233280.$$

- (433211) There are six possible partitions.

The partition can be $3 + 3 + 2 + 411$, where the labels in 411 must be consecutive and the order does not matter. There are $7 \times 3 \times \binom{6}{2} \times 4 = 1260$ ways to choose the four groups.

The partition can be $3 + 3 + 1421$, where the labels in 1421 must be consecutive and the order can be 1421, 1241. There are $6 \times 2 \times \binom{5}{2} = 120$ ways to choose the three groups.

The partition can be $2 + 13431$. There are $5 \times 4 = 20$ ways to choose the two groups.

The partition can be $3 + 11211$ with a 1 being replaced by 3 and a 1 being replaced by 4. There are $5 \times 4 \times 3 \times 4 = 240$ ways to choose the two groups.

However, the partition $3 + 31241$ are counted twice, where the labels in 31241 must be consecutive and the order can be 31241, 14213. There are $5 \times 2 \times 4 = 40$ ways to choose the two groups.

The partition can be $3 + 22311$ with a 2 being replaced by 4, where the labels in 22311 must be consecutive and the order can be 22311, 12321, 11322. There are $5 \times 2 \times 3 \times 4 = 120$ ways to choose the two groups.

The order can be 233211, 123321, 112332 with a 2 being replaced by 4. There are $4 \times 2 \times 3 = 24$ ways to choose this group.

However, the orders 233411, 114332 are counted twice. There are $4 \times 2 = 8$ ways to choose this group.

The number of ways in this case is

$$(1260 + 120 + 20 + 240 - 40 + 120 + 24 - 8) \times \binom{4}{1}^2 \times \binom{4}{2} \times \binom{4}{3}^2 = 2666496.$$

- (4331111) There are four possible partitions.

The partition can be $3 + 3 + 11411$, where the labels in 11411 must be consecutive. There are $5 \times \binom{4}{2} = 30$ ways to choose the three groups.

The partition can be $3 + 111 + 111$ with a 1 being replaced by 3 and a 1 being replaced by 4. There are $10 \times 6 \times 5 \times 3 = 900$ ways to choose the three groups.

However, the partition $3 + 311411$ is counted twice, where the labels in

311411 must be consecutive and the order can be 311411, 114113. There are $4 \times 2 \times 3 = 24$ ways to choose the two groups.

The partition 3113 + 111 with a 1 being replaced by 4 is counted twice, where the labels in each group are in these consecutive orders. There are $12 \times 5 = 60$ ways to choose the two groups.

The orders 3411113, 3141113, 3111413, 3111143 are counted twice. There are $3 \times 4 = 12$ ways to choose this group.

The partition can be 4331 + 111, where the labels in each group must be consecutive and the order in 4331 can be 4331, 1334. There are $12 \times 2 = 24$ ways to choose the two groups.

The order can be 1134311. There are 3 ways to choose this group.

The number of ways in this case is

$$(30 + 900 - 24 - 60 - 12 + 24 + 3) \times \binom{4}{1}^4 \times \binom{4}{3}^2 = 3526656.$$

– (432221) There are six possible partitions.

The partition can be 3+2+4221, where the labels in 4221 must be consecutive and the order can be 4221, 1224. There are $6 \times 2 \times 5 \times 4 = 240$ ways to choose the three groups.

The partition can be 2+22311 with a 1 being replaced by 4, where the labels in 22311 must be consecutive and the order can be 22311, 12321, 11322. There are $5 \times 2 \times 3 \times 4 = 120$ ways to choose the two groups.

The partition can be 2+23421 where the labels in 23421 must be consecutive and the order can be 23421, 12432. There are $5 \times 2 \times 4 = 40$ ways to choose the two groups.

The partition can be 222 + 431, where the labels in each group must be consecutive and the order does not matter. There are $10 \times 2 \times 3! = 120$ ways to choose the two groups.

However, the orders 413222, 222341, 222314, 143222 are counted twice. There are $4 \times 4 = 16$ ways to choose this group.

The order can be 122211, 112221 with a 1 being replaced by 3 and a 1 being replaced by 4. There are $4 \times 3 \times 2 \times 2 = 48$ ways to choose this group.

The order can be 223221, 122322 with a 2 being replaced by 4. There are $4 \times 4 \times 2 = 32$ ways to choose this group.

However, the orders 223421, 124322 are counted twice. There are $4 \times 2 = 8$ ways to choose this group.

The number of ways in this case is

$$(240 + 120 + 40 + 120 - 16 + 48 + 32 - 8) \times \binom{4}{1} \times \binom{4}{2}^3 \times \binom{4}{3} = 1990656.$$

– (4322111) There are six possible partitions.

The partition can be $3 + 2 + 11211$ with a 1 being replaced by 4. There are $5 \times 4 \times 4 \times 3 = 240$ ways to choose the three groups.

The partition can be $3 + 422 + 111$, where the labels in each group must be consecutive and the order does not matter. There are $10 \times 2 \times 3 \times 3 = 180$ ways to choose the three groups.

The partition can be $3 + 122211$ with a 2 being replaced by 4, where the labels in 122211 must be consecutive and the order can be 122211, 112221. There are $4 \times 3 \times 2 \times 3 = 72$ ways to choose the two groups.

The partition can be $1221 + 111$ with a 1 being replaced by 3 and a 1 being replaced by 4. There are $12 \times 5 \times 4 = 240$ ways to choose the two groups.

However, the partition $4223 + 111$ is counted twice, where the labels in each group must be consecutive and the order in 4223 can be 4223, 3224. There are $12 \times 2 = 24$ ways to choose the two groups.

The orders 4221113, 3114221, 3111224, 1224113 are counted twice. There are $3 \times 4 = 12$ ways to choose this group.

The order can be 2231211, 1232211, 1122321, 1121322 with a 2 being replaced by 4. There are $3 \times 3 \times 4 = 36$ ways to choose this group.

The order can be 1121211 with a 1 being replaced by 3 and a 1 being replaced by 4. There are $3 \times 5 \times 4 = 60$ ways to choose this group.

However, the orders 3124211, 1124213 are counted twice. There are $3 \times 2 = 6$ ways to choose this group.

The number of ways in this case is

$$(240 + 180 + 72 + 240 - 24 - 12 + 36 + 60 - 6) \times \binom{4}{1}^3 \times \binom{4}{2}^2 \times \binom{4}{3} = 7243776.$$

- (43211111) There are five possible partitions.

The partition can be $3 + 2 + 111 + 111$ with a 1 being replaced by 4. There are $10 \times 6 \times 3 \times 2 = 360$ ways to choose the four groups.

The partition can be $3 + 1221 + 111$ with a 2 being replaced by 4. There are $12 \times 2 \times 2 = 48$ ways to choose the three groups.

The partition can be $3 + 1121211$ with a 2 being replaced by 4. There are $3 \times 2 \times 2 = 12$ ways to choose the two groups.

The partition can be $22311 + 111$ with a 2 being replaced by 4, where the labels in each group must be consecutive and the order in 22311 can be 22311, 12321, 11322. There are $6 \times 2 \times 3 = 36$ ways to choose the two groups.

The partition can be $11211 + 111$ with a 1 being replaced by 3 and a 1 being replaced by 4. There are $6 \times 7 \times 6 = 252$ ways to choose the two groups.

However, the partition $31241 + 111$ is counted twice, where the labels in each group must be consecutive and the order in 31241 can be 31241, 14213. There are $6 \times 2 = 12$ ways to choose the two groups.

The orders 31141211, 31111241, 14211113, 11214113 are counted twice. So there are $2 \times 4 = 8$ ways to choose this group.

The number of ways in this case is

$$(360 + 48 + 12 + 36 + 252 - 12 - 8) \times \binom{4}{1}^5 \times \binom{4}{2} \times \binom{4}{3} = 16908288.$$

- (431111111) The order must be 111111111 with a 1 being replaced by 3 and a 1 being replaced by 4. The number of ways is $9 \times 8 \times \binom{4}{1}^7 \times \binom{4}{3} = 4718592$.

- (422222) There are two possible partitions.

The partition can be $222 + 222$ with a 2 being replaced by 4. There are $10 \times 6 = 60$ ways to choose the two groups.

The partition can be $2 + 22422$, where the labels in 22422 are in this consecutive order. There are $5 \times 4 = 20$ ways to choose the two groups.

However, the orders 224222, 222422 are counted twice. There are $4 \times 2 = 8$ ways to choose the this group.

The number of ways in this case is $(60 + 20 - 8) \times \binom{4}{2}^5 = 559872$.

- (4222211) There are four possible partitions.

The partition can be $2 + 222 + 411$, where the labels in each group must be consecutive and the order does not matter. There are $10 \times 2 \times 3 \times 3 = 180$ ways to choose the three groups.

However, the partition $2222 + 411$ is counted twice, where the labels in each group must be consecutive and the order does not matter. There are $12 \times 3 = 36$ ways to choose the two groups.

The partition can be $2 + 122211$ with a 1 being replaced by 4, where the labels in 122211 must be consecutive and the order can be 122211, 112221. There are $4 \times 3 \times 2 \times 3 = 72$ ways to choose the two groups.

The partition can be $1221 + 222$ with a 2 being replaced by 4. There are $12 \times 5 = 60$ ways to choose the two groups.

The order can be 1222221 with a 2 being replaced by 4. There are $3 \times 5 = 15$ ways to choose this group.

The number of ways in this case is

$$(180 - 36 + 72 + 60 + 15) \times \binom{4}{1}^2 \times \binom{4}{2}^4 = 6034176.$$

- (42221111) There are five possible partitions.

The partition can be $2 + 1221 + 111$ with a 1 being replaced by 4. There are $12 \times 5 \times 2 = 120$ ways to choose the three groups.

The partition can be $2 + 1121211$ with a 1 being replaced by 4. There are $3 \times 5 \times 2 = 30$ ways to choose the two groups.

The partition can be $11211 + 222$ with a 2 being replaced by 4. There are $6 \times 4 = 24$ ways to choose the two groups.

The partition can be $1221 + 1221$ with a 2 being replaced by 4. There are $3 \times 4 = 12$ ways to choose the two groups.

The order can be 12221211, 11222211, 11212221 with a 2 being replaced by 4. There are $2 \times 4 \times 3 = 24$ ways to choose this group.

The number of ways in this case is

$$(120 + 30 + 24 + 12 + 24) \times \binom{4}{1}^4 \times \binom{4}{2}^3 = 11612160.$$

- (422111111) There are four possible partitions.

The two 2's and the 4 are in consecutive positions and the order does not matter. For example, *abbbbccdefghi* can be partitioned as $\{abc, bcd, def, ghi, bb\}$. There are $7 \times 3 = 21$ ways to choose their positions.

The partition can be $2 + 11211 + 111$ with a 1 being replaced by 4. There are $3! \times 7 = 42$ ways to choose the three groups.

The partition can be $11211 + 1221$ with a 2 being replaced by 4. There are $2 \times 3 = 6$ ways to choose the two groups.

The order can be 112121211 with a 2 being replaced by 4. There are 3 ways to choose this group.

The number of ways in this case is $(21 + 42 + 6 + 3) \times \binom{4}{1}^6 \times \binom{4}{2}^2 = 10616832$.

- (33332) The labels can be arbitrary as we can partition this case as $3 + 3 + 3 + 3 + 2$. The number of ways is $\binom{9}{4} \times 5 \times \binom{4}{2} \times \binom{4}{3}^4 = 967680$.

- (333311) There are two possible partitions.

The partition can be $3 + 3 + 3 + 311$, where the labels in 311 must be consecutive and the order does not matter. There are $7 \times 3 \times \binom{6}{3} = 420$ ways to choose the four groups.

However, the partition $3 + 3 + 3113$ is counted twice, where the labels in 3113 are in this consecutive order. There are $6 \times \binom{5}{2} = 60$ ways to choose the three groups.

The order can be $133311, 113331$ with a 1 being replaced by 3. There are $4 \times 3 \times 2 = 24$ ways to choose this group.

However, the orders $333311, 133331, 113333$ are counted twice. There are $4 \times 3 = 12$ ways to choose this group.

The number of ways in this case is $(420 - 60 + 24 - 12) \times \binom{4}{1}^2 \times \binom{4}{3}^4 = 1523712$.

- (333221) There are four possible partitions.

The partition can be $3 + 3 + 3221$, where the labels in 3221 must be consecutive and the order can be $3221, 1223$. There are $6 \times 2 \times \binom{5}{2} = 120$ ways to choose the three groups.

The partition can be 3+2+2331, where the labels in 2331 must be consecutive and the order can be 2331, 1332. There are $6 \times 2 \times 5 \times 4 = 240$ ways to choose the three groups.

The partition can be 3+22311 with a 1 being replaced by 3, where the labels in 22311 must be consecutive and the order can be 22311, 12321, 11322. There are $5 \times 2 \times 3 \times 4 = 120$ ways to choose the two groups.

However, the partition 3+22331 is counted twice, where the labels in 22331 must be consecutive and the order can be 22331, 13322. Note that there are $5 \times 2 \times 4 = 40$ ways to choose the two groups.

The order can be 233211, 123321, 112332 with a 1 being replaced by 3. There are $4 \times 2 \times 3 = 24$ ways to choose this group.

The number of ways in this case is

$$(120 + 240 + 120 - 40 + 24) \times \binom{4}{1} \times \binom{4}{2}^2 \times \binom{4}{3}^3 = 4276224.$$

– (3332111) There are five possible partitions.

The partition can be 3 + 3 + 3 + 2 + 111, where the labels in 111 must be consecutive. There are $7 \times \binom{6}{3} \times 3 = 420$ ways to choose the five groups.

The partition can be 3 + 3 + 11211 with a 1 being replaced by 3. There are $5 \times 4 \times \binom{4}{2} = 120$ ways to choose the three groups.

The partition can be 2 + 133311, where the labels in 133311 must be consecutive and the order can be 133311, 113331. There are $4 \times 2 \times 3 = 24$ ways to choose the two groups.

The partition can be 2331 + 311, where the labels in each group must be consecutive and the order in 2331 can be 2331, 1332 while the order in 311 does not matter. There are $12 \times 2 \times 3 = 72$ ways to choose the two groups.

However, the orders 3111332, 2331113 are counted twice. There are $3 \times 2 = 6$ ways to choose this group.

The order can be 3132311, 1132313. For example, *aaabcccddeefg* can be partitioned as $\{abc, cde, cde, efg, aa\}$. There are $3 \times 2 = 6$ ways to choose this group.

The number of ways in this case is

$$(420 + 120 + 24 + 72 - 6 + 6) \times \binom{4}{1}^3 \times \binom{4}{2} \times \binom{4}{3}^3 = 15630336.$$

- (33311111) The partition must be $3 + 3 + 311 + 111$, where the labels in each group must be consecutive and the order does not matter. There are $10 \times 6 \times \binom{3}{2} = 180$ ways to choose the four groups.

However, the partition $3 + 3113 + 111$ is counted twice, where the labels in each group are in these consecutive orders. There are $12 \times 2 = 24$ ways to choose the three groups.

The partition $3 + 3111113$ is counted twice, where the labels in 3111113 are in this consecutive order. There are $3 \times 2 = 6$ ways to choose the two groups.

The number of ways in this case is $(180 - 24 - 6) \times \binom{4}{1}^5 \times \binom{4}{3}^3 = 9830400$.

- (332222) There are two possible partitions.

The partition can be $3 + 3 + 2 + 222$, where the labels in 222 must be consecutive. There are $7 \times \binom{6}{2} \times 4 = 420$ ways to choose the four groups.

However, the partition $3 + 3 + 2222$ is counted twice, where the labels in 2222 must be consecutive. There are $6 \times \binom{5}{2} = 60$ ways to choose the three groups.

The order can be $322322, 223223$. There are $4 \times 2 = 8$ ways to choose this group.

The number of ways in this case is $(420 - 60 + 8) \times \binom{4}{2}^4 \times \binom{4}{3}^2 = 7630848$.

- (3322211) There are six possible partitions.

The partition can be $3 + 3 + 2 + 1221$, where the labels in 1221 are in this consecutive order. There are $6 \times \binom{5}{2} \times 3 = 180$ ways to choose the four groups.

The partition can be $3 + 2 + 22311$, where the labels in 22311 must be consecutive and the order can be $22311, 12321, 11322$. Thus, there are $5 \times 3 \times 4 \times 3 = 180$ ways to choose the three groups.

The partition can be $3 + 222 + 311$, where the labels in each group must be consecutive and the order does not matter. There are $10 \times 2 \times 3 \times 3 = 180$ ways to choose the three groups.

However, the partition $3 + 222311$ is counted twice, where the labels in

222311 must be consecutive and the order can be 222311, 113222. There are $4 \times 2 \times 3 = 24$ ways to choose the two groups.

The partition 3113+222 is counted twice, where the labels in each group are in these consecutive orders. There are 12 ways to choose the two groups.

The partition can be 2+233211, where the labels in 233211 must be consecutive and the order can be 233211, 123321, 112332. There are $4 \times 3 \times 3 = 36$ ways to choose the two groups.

The partition can be 3+122211 with a 1 being replaced by 3, where the labels in 122211 must be consecutive and the order can be 122211, 112221. There are $4 \times 3 \times 2 \times 3 = 72$ ways to choose the two groups.

However, the order 3222113, 3122213, 3112223 is counted twice. There are $3 \times 3 = 9$ ways to choose this group.

The order can be 2231211, 1232211, 1122321, 1121322 with a 1 being replaced by 3. There are $3 \times 3 \times 4 = 36$ ways to choose this group.

However, the orders 2233211, 1123322 are counted twice. There are $3 \times 2 = 6$ ways to choose this group.

The number of ways in this case is

$$(180 + 180 + 180 - 24 - 12 + 36 + 72 - 9 + 36 - 6) \times \binom{4}{1}^2 \times \binom{4}{2}^3 \times \binom{4}{3}^2 = 35002368.$$

– (33221111) There are six possible partitions.

The partition can be 3+3+2+11211, where the labels in 11211 are in this consecutive order. There are $5 \times \binom{4}{2} \times 2 = 60$ ways to choose the four groups.

The partition can be 2+2331+111, where the labels in each group must be consecutive and the order in 2331 can be 2331, 1332. There are $12 \times 2 \times 2 = 48$ ways to choose the three groups.

The partition can be 3+1221+111 with a 1 being replaced by 3. There are $12 \times 5 \times 2 = 120$ ways to choose the three groups.

However, the partition 1221+3113 is counted twice, where the labels in each group are in these consecutive orders. There are $3 \times 2 = 6$ ways to choose the two groups.

The partition can be 2+1132311, where the labels in 1132311 are in this consecutive order. There are $3 \times 2 = 6$ ways to choose the two groups.

The partition can be $3 + 1121211$ with a 1 being replaced by 3. There are $3 \times 5 \times 2 = 30$ ways to choose the two groups.

The partition can be $22311 + 111$ with a 1 being replaced by 3, where the labels in each group must be consecutive and the order in 22311 can be 22311, 12321, 11322. There are $6 \times 5 \times 3 = 90$ ways to choose the two groups.

However, the partition $22331 + 111$ is counted twice, where the labels in each group must be consecutive and the order in 22331 can be 22331, 13322. There are $6 \times 2 = 12$ ways to choose the two groups.

The order 11322311 is counted twice. There are 2 ways to choose this group.

The number of ways in this case is

$$(60 + 48 + 120 - 6 + 6 + 30 + 90 - 12 - 2) \times \binom{4}{1}^4 \times \binom{4}{2}^2 \times \binom{4}{3}^2 = 49250304.$$

– (332111111) There are two possible partitions.

The partition can be $3 + 3 + 2 + 111 + 111$, where the labels in 111 must be consecutive. There are $\binom{5}{2} \times 3 = 30$ ways to choose the five groups.

The partition can be $3 + 11211 + 111$ with a 1 being replaced by 3. There are $3! \times 7 = 42$ ways to choose the three groups.

However, the partition $3113 + 11211$ is counted twice, where the labels in each group are in these consecutive orders. There are 2 ways to choose the two groups.

The number of ways in this case is $(30 + 42 - 2) \times \binom{4}{1}^6 \times \binom{4}{2} \times \binom{4}{3}^2 = 27525120$.

– (3222221) There are three possible partitions.

The partition can be $2 + 223221$, where the labels in 223221 must be consecutive and the order can be 223221, 122322. There are $4 \times 2 \times 3 = 24$ ways to choose the two groups.

The partition can be $3221 + 222$, where the labels in each group must be consecutive and the order in 3221 can be 3221, 1223. There are $12 \times 2 = 24$ ways to choose the two groups.

However, the orders 2223221, 1223222 are counted twice. There are $3 \times 2 = 6$ ways to choose this group.

The order can be 1222221 with a 1 being replaced by 3. There are $3 \times 2 = 6$ ways to choose this group.

The number of ways in this case is $(24+24-6+6) \times \binom{4}{1} \times \binom{4}{2}^5 \times \binom{4}{3} = 5971968$.

- (32222111) There are six possible partitions.

The partition can be $3 + 2 + 222 + 111$, where the labels in each group must be consecutive. There are $10 \times 2 \times 3 \times 2 = 120$ ways to choose the four groups.

However, the partition $3 + 2222 + 111$ is counted twice, where the labels in each group must be consecutive. There are $12 \times 2 = 24$ ways to choose the three groups.

The partition can be $3 + 2 + 122211$, where the labels in 122211 must be consecutive and the order can be $122211, 112221$. There are $4 \times 2 \times 3 \times 2 = 48$ ways to choose the three groups.

The partition can be $2 + 2231211$, where the labels in 2231211 must be consecutive and the order can be $2231211, 1232211, 1122321, 1121322$. There are $3 \times 4 \times 2 = 24$ ways to choose the two groups.

The partition can be $11211 + 222$ with a 1 being replaced by 3. There are $6 \times 4 = 24$ ways to choose the two groups.

However, the orders $22231211, 11213222$ are counted twice. There are $2 \times 2 = 4$ ways to choose this group.

The partition can be $1221 + 1221$ with a 1 being replaced by 3, where the labels in each group are in this consecutive order. There are $3 \times 4 = 12$ ways to choose the two groups.

The order can be $12221211, 11222211, 11212221$ with a 1 being replaced by 3. There are $2 \times 4 \times 3 = 24$ ways to choose this group.

The number of ways in this case is

$$(120 - 24 + 48 + 24 + 24 - 4 + 12 + 24) \times \binom{4}{1}^3 \times \binom{4}{2}^4 \times \binom{4}{3} = 74317824.$$

- (322211111) There are six possible partitions.

The three 2's are in consecutive positions. For example, $abccddeefffghi$ can be partitioned as $\{abc, cde, def, ghi, ff\}$. There are $7 \times 6 = 42$ ways to choose their positions.

The partition can be $3 + 2 + 1221 + 111$, where the labels in each group are in these consecutive orders. There are $4! = 24$ ways to choose the four groups.

The partition can be $3 + 2 + 1121211$, where the labels in 1121211 are in this consecutive order. There are $3! = 6$ ways to choose the three groups.

The partition can be $2 + 22311 + 111$, where the labels in each group must be consecutive and the order in 22311 can be $22311, 12321, 11322$. There are $3! \times 3 = 18$ ways to choose the three groups.

However, the partition $222311 + 111$ is counted twice, where the labels in each group must be consecutive and the order in 222311 can be $222311, 113222$. There are $2 \times 2 = 4$ ways to choose the two groups.

The partition can be $11211 + 1221$ with a 1 being replaced by 3, where the labels in each group are in these consecutive orders. There are $2 \times 6 = 12$ ways to choose the two groups.

The order can be 112121211 with a 1 being replaced by 3. There are 6 ways to choose this group.

The number of ways in this case is

$$(42 + 24 + 6 + 18 - 4 + 12 + 6) \times \binom{4}{1}^5 \times \binom{4}{2}^3 \times \binom{4}{3} = 92012544.$$

- (2222222) The partition must be $2 + 222 + 222$, where the labels in 222 must be consecutive. There are $10 \times 3 = 30$ ways to choose the three groups.

However, the partition $2222 + 222$ is counted twice, where the labels in each group must be consecutive. There are 12 ways to choose the two groups.

The number of ways is $(30 - 12) \times \binom{4}{2}^7 = 5038848$.

- (22222211) There are two possible partitions.

The partition can be $2 + 1221 + 222$, where the labels in each group are in these consecutive orders. There are $12 \times 2 = 24$ ways to choose the three groups.

However, the partition $2222 + 1221$ is counted twice, where the labels in each group are in these consecutive orders. There are 6 ways to choose the two groups.

The partition can be $2 + 1222221$. There are $3 \times 2 = 6$ ways to choose the two groups.

The number of ways in this case is $(24 - 6 + 6) \times \binom{4}{1}^2 \times \binom{4}{2}^6 = 17915904$.

- (222221111) There are three possible partitions.

The partition can be $2 + 11211 + 222$, where the labels in each group are in these consecutive orders. There are $3! = 6$ ways to choose the three groups.

However, the partition $2222 + 11211$ is counted twice, where the labels in each group are in these consecutive orders. There are 2 ways to choose the two groups.

The partition can be $2 + 1221 + 1221$, where the labels in 1221 are in this consecutive order. There are 3 ways to choose the three groups.

The partition can be $2 + 12221211$, where the labels in 12221211 must be consecutive and the order can be 12221211, 11222211, 11212221. There are $2 \times 3 = 6$ ways to choose the two groups.

The number of ways in this case is $(6 - 2 + 3 + 6) \times \binom{4}{1}^4 \times \binom{4}{2}^5 = 25878528$.

The total number of ways to choose 4 nice triples with an equal pair is

$$\begin{aligned}
& 324 + 480 + 2016 + 81600 + 165888 + 15120 + 207360 + 912384 + 163840 \\
& + 161280 + 233280 + 2666496 + 3526656 + 1990656 + 7243776 + 16908288 \\
& + 4718592 + 559872 + 6034176 + 11612160 + 10616832 + 967680 + 1523712 \\
& + 4276224 + 15630336 + 9830400 + 7630848 + 35002368 + 49250304 + 27525120 \\
& + 5971968 + 74317824 + 92012544 + 5038848 + 17915904 + 25878528 \\
& = 440593684.
\end{aligned}$$

Now, we are able to find the number of ways to win the game under condition (i). We list out all possible cases as follows. For example, the case (5, 3, 0, 6) means there are 5, 3, 0, 6 cards of types A, B, C, D respectively.

- (14, 0, 0, 0): 440593684 ways
- (0, 0, 0, 14): 161280 ways
- (12, 2, 0, 0): $40399783 \times 54 = 2181588282$ ways
- (12, 0, 0, 2): $40399783 \times 42 = 1696790886$ ways
- (2, 0, 0, 12): $54 \times 8960 = 483840$ ways
- (11, 3, 0, 0): $47037380 \times 484 = 22766091920$ ways
- (11, 0, 0, 3): $47037380 \times 28 = 1317046640$ ways
- (3, 0, 0, 11): $484 \times 53760 = 26019840$ ways
- (9, 5, 0, 0): $2742868 \times 19200 = 52663065600$ ways

- (9, 0, 0, 5): $2742868 \times 1008 = 2764810944$ ways
- (5, 0, 0, 9): $19200 \times 2240 = 43008000$ ways
- (9, 3, 2, 0): $2742868 \times 484 \times 54 = 71687598048$ ways
- (9, 3, 0, 2): $2742868 \times 484 \times 42 = 55757020704$ ways
- (9, 2, 0, 3): $2742868 \times 54 \times 28 = 4147216416$ ways
- (3, 2, 0, 9): $484 \times 54 \times 2240 = 58544640$ ways
- (8, 6, 0, 0): $1748756 \times 65272 = 114144801632$ ways
- (8, 0, 0, 6): $1748756 \times 336 = 587582016$ ways
- (6, 0, 0, 8): $65272 \times 10080 = 657941760$ ways
- (8, 3, 3, 0): $1748756 \times 484 \times 484 = 409656585536$ ways
- (8, 3, 0, 3): $1748756 \times 484 \times 28 = 23699141312$ ways
- (3, 3, 0, 8): $484 \times 484 \times 10080 = 2361300480$ ways
- (6, 6, 2, 0): $65272 \times 65272 \times 54 = 230063435136$ ways
- (6, 6, 0, 2): $65272 \times 65272 \times 42 = 178938227328$ ways
- (6, 2, 0, 6): $65272 \times 54 \times 336 = 1184295168$ ways
- (6, 5, 3, 0): $65272 \times 19200 \times 484 = 606559641600$ ways
- (6, 5, 0, 3): $65272 \times 19200 \times 28 = 35090227200$ ways
- (6, 3, 0, 5): $65272 \times 484 \times 1008 = 31844381184$ ways
- (5, 3, 0, 6): $19200 \times 484 \times 336 = 3122380800$ ways
- (6, 3, 3, 2): $65272 \times 484 \times 484 \times 42 = 642195020544$ ways
- (6, 3, 2, 3): $65272 \times 484 \times 54 \times 28 = 47766571776$ ways
- (3, 3, 2, 6): $484 \times 484 \times 54 \times 336 = 4250340864$ ways
- (5, 3, 3, 3): $19200 \times 484 \times 484 \times 28 = 125936025600$ ways
- (3, 3, 3, 5): $484 \times 484 \times 484 \times 1008 = 114286943232$ ways

Since types A, B, C are symmetric, the total number of cases satisfying condition (i) is

$$\begin{aligned}
& 6 \times (2181588282 + 22766091920 + 52663065600 + 71687598048 + 55757020704 \\
& + 4147216416 + 58544640 + 114144801632 + 23699141312 + 1184295168 \\
& + 606559641600 + 35090227200 + 31844381184 + 3122380800 + 47766571776)
\end{aligned}$$

$$\begin{aligned}
& + 3 \times (440593684 + 1696790886 + 483840 + 1317046640 + 26019840 + 2764810944 \\
& + 43008000 + 587582016 + 657941760 + 409656585536 + 2361300480 + 230063435136 \\
& + 178938227328 + 642195020544 + 4250340864 + 125936025600) + 161280 \\
& + 114286943232 \\
& = 11353128141498.
\end{aligned}$$

Finally, to satisfy condition (ii), we have to pick the cards with labels

$$1, 9, 11, 19, 21, 29, 31, 41, 51, 61, 71, 81, 91$$

plus one more card with one of these labels. Clearly, there are

$$13 \times \binom{4}{1}^{12} \times \binom{4}{2} = 1308622848$$

ways to do so.

Therefore, the probability of winning the game is

$$\frac{11353128141498 + 1308622848}{\binom{136}{14}} = \frac{11354436764346}{\binom{136}{14}}.$$

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